

Open Source Robotics

Multi-Robot Simulators



From left to right: Pioneer robots and the Foot-bot robot

This is the second article in a series that focuses on open source software for robotics. This part introduces the Linux enthusiast to multi-robot simulators, by taking a peek into Stage and ARGoS.

Part—2

A robot enthusiast's flight of fancy could envisage a futuristic society where robots and human beings coexist. This apparent utopia would be cohabited by human beings and robots that obey human beings and help in their daily chores, do the jobs that are redundant, come to rescue when human lives are in jeopardy, and are intelligent enough to protect the human populace.

The development of paradigms for a human-robot society led to science fiction writer Isaac Asimov coining the Three Laws of Robotics:

- I. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
- II. A robot must obey the orders given to it by human beings, except where such orders would conflict with the First Law.
- III. A robot must protect its own existence, as long as such protection does not conflict with the First or Second Laws.

These laws have come to form the basis of a stable society containing robots that are subservient to the human population.

Multi-robot systems

To make prophecies about such futuristic societies, it is important to study multi-robot interactions. It is easy to see that a multi-robot team will have clear advantages over a

single robot system. Also, improved performance due to distributed sensing, enhanced fault tolerance and parallelism, add to the virtues of a robotic team.

Like many other emerging fields of robotics, algorithms for multi-robot systems are inspired by nature. Foraging for food, division of labour, nest-building, cumulative defence against enemies, leader following, flocking, etc, are common in human, insect and animal societies.

A unique feature of multi-robot systems is that one can distinguish between individual-level behaviour and team-level behaviour. The former refers to what an individual in a team does. The latter refers to the behaviour of the team as a whole. For instance, in schools of fish, the individual behaviour of each fish is to mainly avoid collision with neighbours, and match their attitude. The team-level behaviour, on the other hand, is a remarkable display of cohesive and coordinated motion.

The study of the relationship between individual and team-level behaviour is just beginning. It's a long road to constructing multi-robot systems that are able to display the same complexity and performance of natural systems like insect societies. Also, the cost for this type of research is prohibitive. Robots are expensive, and producing large quantities of them is out of the reach of today's research